SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I __P.Sunayana(192424303)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

P. Sunayana.
192424303.

CO 1 - Assessment Tool 1

1) Event communication Security - Caesar cipher.

a) Reconstruct the encrypted message using a shift of 4.

Original message : MEET AT NOON.

Using Ceasar cipher, shift each letter 4 positions forward in the alphabet.

M → Q
E → I
E → I
T → X
A → E

T → X
A → E
T → X
N → R
O → S
O → S
N → R.

Encrypted message :- QIIX EX RSSR.

b) How can an authorized member retrieve the original message?

The authorized member (decrypts) decrypts the message by shifting each letter 4 positions backward in the alphabet.

Example :- Q → M ; I → E ; I → E ; X → T.

Thus, the original message is recovered.

Original message :- MEET AT NOON.

encrypted message :- QIIX EX RSSR.

2) Intercepted suspicious Message - Substitution cipher.

a) Deduce the most likely original message.

Intercepted message :- GSRH RH Z HVXING.

using the general monoalphabetic cipher;

where A → Z, B → Y, C → X, and so on:-

so,

Original message :- THIS IS SECRET.

b) Justify your answer by analyzing repeating patterns, letter structures, (or) known cipher characteristics.

→ The message uses a consistent letter - substitution pattern (general monoalphabetic substitution cipher).

→ the substitution technique replaces letter with its reverse - alphabet counterpart.

→ For example :- G → T, S → H

R → I, H → S.

→ The decoded words form a meaningful English sentence.

Eg:- THIS IS SECRET.

→ Therefore, the message was most likely encrypted using monoalphabetic substitution cipher.

3) Secure Messaging system – Vigenere cipher.

a) Demostrate the encryption process step-by-step

plaintext :- HELLO

key : KEY.

7	4	11	11	14
H	E	L	L	O
K	E	Y	K	E
10	4	24	10	4

Encryption :-

$$C_i = P_i + k_i \pmod{26}$$

$C_H = 7 + 10 \bmod 26$

$= 17$ (R)

$C_E = 4 + 4 \bmod 26$

$= 8$ (I)

$C_L = 11 + 24 \bmod 26$

$= 35 \bmod 26 = 9$ (J)

$C_L = 11 + 10 \bmod 26$

$= 21$ (V)

$C_O = 14 + 4 \bmod 26$

$= 18$ (S).

Cipher text :- RIJVS.

b) How does the repeating keyword influence each character?

→ Each plaintext letter is combined with corresponding key letter.

→ Different key letters produce different shifts.

→ The keyword repeats when the message is longer than the key.

→ This makes the cipher more complex than ceasar cipher.

4) Suspicious Digital Asset Investigation - Steganography:

a) How can investigators detect hidden data in an image?

→ Examine the image's metadata for suspicious information.
→ Compare the file size with that of similar images.
→ Analyze pixel values and least significant bits (LSB's).
→ Use steganography detection tools of identify usual patterns.
→ Extract suspected hidden data using appropriate forensic tools.

b) Recommend two techniques to strengthen security.

1) Encryption before embedding:-

Encrypt the secret message before hiding it in the image, so extracted data remains unreadable without the key.

2) Randomized embedding:- Use a secret key to select pixel locations instead of embedding data in predictable positions, making detection and extraction more difficult.

5) Legacy Encryption in corporate communication - Transposition cipher.

Recieved message :- WERRLTEERDSOEEFEAOCAIVDEN.

a) Determine the original message by reversing the encryption.

The ciphertext appears due to use a columnar transposition cipher. however, the exact arrangement of the three levels is not specified clearly though to determine a unique decryption with certainity.

A common transposition example is :-

→ Original message :- WE ARE DECIDED FLEE AT ONCE.

→ Remove spaces :- WEAREDECIDEDFLEEATONCE.

→ Rearrange the letters according to the specified transposi-tion pattern to produce the ciphertext.

b) Why is this method vulnerable in modern communication system ?

→ It rearranges letters without changing them.

→ letter frequencies remain the same as in the plaintext.

→ Repeated patterns can help attackers identify the message.

→ It does not provide modern cryptographic security.

→ It is vulnerable to cryptoanalysis, especially when the arrangement is known (or) guessed.

c) propose one practical enhancement.

Replace the transposition cipher with AES (Advanced Encryption Standard) using a secure mode such as AES-

GCM. It provides strong confidentiality and also detects unauthorized modification of the encrypted message.